

A11 Thesis Proposal First Draft

Designing with Latent Space: Using a Semantic Canvas and an AI Agent to Broaden Exploration and Clarify Intent in Early-Stage Footwear Design

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Abstract

This project studies how a designer-facing “semantic canvas” and a lightweight AI agent can support early-stage industrial design, with footwear as the testbed. The problem is that today’s text-to-image tools often encourage narrow, prompt-driven iteration that can lead to design fixation and make it hard to keep design intent clear as exploration unfolds (Jansson & Smith, 1991; Sio et al., 2015). I propose a two-layer approach: a 2D/3D latent-space canvas that organizes images by semantic similarity, and an event-driven agent that suggests under-explored regions, candidate axes, and seed prompts based on a compact digest of current work (Radford et al., 2021; Härkönen et al., 2020; Shen et al., 2020).

My methods include high-fidelity prototyping and a within-subject user study ($N \approx 10-20$) comparing three conditions: (A) baseline prompt-and-batch, (B) semantic canvas without agent, and (C) semantic canvas with agent. Measures focus on two success criteria: avoiding fixation (breadth, switching behavior, coverage) and clarifying intent (self-report clarity, traceable rationale, alignment between brief and selected directions). Early pilot interviews with two footwear designers affirmed the direction and informed task design; full IRB approval will precede formal studies. Anticipated contributions include a practical interaction pattern for agent-supported exploration, lightweight metrics for breadth/depth of search, and guidance for integrating generative models into design workflows beyond “more outputs,” emphasizing legibility and intent.

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Introduction

Early-stage work with text-to-image tools often follows a prompt-and-grid loop that narrows exploration and allows intent to drift from the brief. In industrial design, and footwear in particular, the first hour matters: early choices shape downstream cost, timing, and feasible directions. My project reframes this phase from “generate more” to “navigate and reason.” I propose a two-layer approach. First, a semantic canvas places results by meaning and exposes axes, clusters, gaps, and simple lineage so designers can see where they are and what remains unexplored. Second, a brief-aware agent reads a compact snapshot of the session and returns a few named, on-canvas suggestions with short reasons. The aim is to reduce fixation by surfacing alternatives and to clarify intent by coupling accepted choices to labels and rationales that remain tied to the brief. The expected outcome is better early decisions and a workflow that is explainable and repeatable.

Background / Literature Review

Prior studies show that early examples can bias later ideas and reduce variety, a pattern widely described as design fixation. This risk is amplified by grid-first interfaces that emphasize immediate visual appeal over structured exploration. In parallel, research on model representations demonstrates that latent spaces contain meaningful directions that can be named and manipulated, and that vision-language embeddings can place images in a shared semantic map. These findings suggest that structure exists and can be surfaced with modest tooling. Yet everyday design tools typically hide this structure; users iterate within narrow neighborhoods without a sense of the larger space. At the same time, text-to-image systems still struggle with compositional reliability, making prompt editing alone a weak steering mechanism. My project addresses this gap by exposing latent organization as a working surface and by adding lightweight, well-timed suggestions that route designers toward viable alternatives while keeping selections tied to the original brief.

Models (Reference & Impact)

The reference model describes the status quo: a brief prompts batches of images; an early appealing result anchors subsequent prompts; the latent space remains hidden; exploration narrows; and intent gradually drifts. The impact model introduces two supports designed to counter these tendencies. The semantic canvas renders the hidden space visible as axes, clusters, gaps, and a simple timeline, enabling designers to locate work, compare options, and notice underexplored regions. The brief-aware agent consumes a compact state description and proposes a small set of named, on-canvas actions with one-line reasons, such as highlighting a sparse but relevant area, suggesting a separating axis, or staging a few samples. Together, the canvas and agent target two criteria that matter in early-stage work: avoiding fixation by widening coverage and branching, and clarifying intent by linking accepted choices to labeled regions and short, persistent explanations.

Methods

I am developing a working prototype and will run a within-subject study that compares three conditions: (C0) baseline prompt-and-grid, (C1) semantic canvas only, and (C2) canvas plus agent. Participants will include 10–20 industrial design students and practicing footwear or product designers. Each participant will complete three short briefs and select several candidates per condition after a brief tutorial. I will measure breadth through coverage across regions of the canvas, visible branching and switching behavior, and the time to a first divergent choice beyond the initial neighborhood. I will measure intent clarity through brief match of accepted picks, time to correct after a poor match, and the use of named axes, tags, and short reasons at acceptance. Experience measures will include time and clicks per accepted pick and short ratings of predictability and legibility. Data will consist of interaction logs, canvas snapshots, short surveys, and brief interviews. Analyses will report within-subject differences with effect sizes and confidence intervals and qualitative coding of fixation and intent-clarification moments. IRB approval will precede recruitment.

Results to Date / Anticipated Findings

The current prototype supports 2D and 3D canvases, semantic placement, and basic overlays, and it integrates an image generation backend. Early demonstrations with two designers indicate that a map-like view helps them see alternatives and that suggestions are most useful when they are named and tied to visible regions. I expect the canvas (C1) to increase breadth relative to baseline (C0) by improving coverage and switching. I further expect the agent (C2) to improve both breadth and intent clarity by prompting exploration of gaps and contrasts and by reviving promising but inactive threads. Known friction points include limits in attribute binding and localized control in current models; the

interface aims to route around these limits by steering exploration toward more reliable areas of the space.

Discussion / Contributions

This work contributes a practical pattern for agent-supported exploration in creative tools. A small state digest from the canvas leads to a few named, on-canvas suggestions with concise reasons that are easy to audit and ignore if not useful. It also contributes a legible interaction model that treats latent space as a working surface rather than a hidden substrate, exposing axes, clusters, gaps, and simple lineage so designers can navigate deliberately. Finally, it offers simple, reusable measures for early-stage evaluation that emphasize breadth and intent clarity over raw output counts. Although the study focuses on footwear, the approach is intended to transfer to other areas of industrial design where early exploration benefits from visible structure and lightweight guidance.

Limitations and Next Steps

The study will involve a modest sample and a single domain, and it focuses on the first hour of work rather than full projects. Findings will also reflect current model limits, including imperfect attribute binding and uneven part-level control. Next steps include completing agent actions for gap highlighting, axis proposals, and thread revival; polishing overlays and logging; running a small pilot to refine task wording and suggestion timing; securing IRB approval; and conducting the full within-subject study. After analysis, I plan to explore integration with an industry partner and to prepare a publication that details the interaction pattern, measures, and results, along with guidance for adapting the approach to adjacent product domains.

References (APA)

Final list in APA 7th; match in-text citations.

Introduction

Text-to-image systems have become common tools for early-stage ideation in industrial design. Their speed and visual immediacy are useful; however, a recurring pattern of **prompt → grid → selection → prompt revision** can narrow exploration and encourage premature commitment to a single direction. Prior research describes this tendency as **design fixation**, in which early examples bias subsequent ideas and reduce variety (Jansson & Smith, 1991; Sio, Kotovsky, & Cagan, 2015). In parallel, the rationale linking selected images to the original brief often remains implicit. As prompts evolve, intent can drift, and later choices are difficult to justify in relation to earlier aims.

Modern models nevertheless encode rich **latent spaces** in which images and concepts occupy structured neighborhoods. Research demonstrates that some of these directions are interpretable and can be named or manipulated as meaningful factors (Härkönen, Hertzmann, Lehtinen, & Paris, 2020; Shen, Gu, Tang, & Zhou, 2020). In many tools this structure is not exposed to designers; outputs are presented as disjointed grids rather than as a map that reveals adjacency, gaps, or alternative routes.

This project studies latent space as a **designer-facing medium** for early-stage work, with footwear as the primary testbed. I propose a two-layer approach: (1) a **semantic canvas** that places results in 2D/3D according to vision-language embeddings and supports axes, clusters, gaps, and simple

lineage; and (2) a **brief-aware agent** that reads a compact state description of the current session and returns a small number of named, on-canvas suggestions with short rationales. The intended outcomes are twofold: **reducing fixation** by making alternatives and under-explored areas visible, and **clarifying intent** by coupling images to named axes, tags, and succinct reasons that remain tied to the brief.

Footwear provides an appropriate domain because it combines accessible datasets, clear part vocabularies (e.g., upper, tongue, toe cap, midsole, outsole), and well-defined functional targets. Insights from this domain are expected to generalize to other areas of industrial design where early exploration benefits from breadth and traceable rationale.

[Figure 1 — Placeholder]

Comparison of a grid loop versus a semantic map with labeled regions and axes.

Background / Literature Review

1. Design fixation and breadth of exploration

Empirical studies in engineering and design consistently report **fixation effects**: exposure to early examples can constrain the range of ideas, promote feature reuse, and limit novelty (Jansson & Smith, 1991). A meta-analysis synthesizing multiple experiments confirms that examples can aid productivity but also reliably increase bias toward seen features, particularly under time pressure and when examples are visually salient (Sio et al., 2015). Within prompt-driven image generation, the most compelling early outputs frequently anchor subsequent prompts and selections, fostering iterative refinement around a single path while suppressing lateral movement. These findings motivate interfaces that support **coverage of alternatives**, **branching**, and **timely switching** rather than prolonged deepening of an initial line.

[Figure 2 — Placeholder]

Line chart contrasting a single deep path with multiple branching paths over time.

2. Legibility and controllable directions in latent space

Work on interpretable latent variables shows that some high-dimensional model spaces can be navigated along **coherent directions**. GANSpace identifies principal directions corresponding to human-understandable changes (Härkönen et al., 2020). InterFaceGAN learns linear boundaries associated with semantic attributes and demonstrates localized, predictable edits (Shen et al., 2020). Complementary efforts align images and text in a joint embedding space (e.g., CLIP), enabling placement and retrieval by semantic similarity (Radford et al., 2021). Collectively, these results indicate that latent organization is not arbitrary; with modest tooling it can be surfaced to users as labeled axes, neighborhoods, or contrasts. Exposing such structure provides a basis for **visual navigation** and for attaching **names and reasons** to movements within the space.

[Figure 3 — Placeholder]

2D scatter of images with two labeled axes, cluster boundaries, and one highlighted gap.

3. Interface mediation in the presence of model limits

Despite recent progress, text-to-image systems continue to show **compositional weaknesses**, such as attribute binding failures and relational errors. Benchmarks like T2I-CompBench document these limitations across open-world prompts (Huang, Sun, Xie, Li, & Liu, 2023). In practice, attempting to overcome such limits through additional prompting alone can exacerbate fixation: designers spend more time editing text while remaining near the same region of the space. An interface that reveals the landscape and proposes **workable detours**—for example, by shifting along a more reliable axis or by sampling adjacent neighborhoods—can mitigate these effects and preserve intent.

4. Role and scope of the agent

The agent in this project is designed to **propose** rather than to automate. Clustering, density estimation, novelty scoring, and simple coverage metrics are computed server-side. The agent consumes a **state digest** (a small canvas thumbnail, compact summaries of salient regions, and a short history of recent actions) and returns a limited set of **on-canvas suggestions** with concise rationales—for instance, highlighting a sparse but relevant area, proposing an axis that separates recent choices, or staging a small number of samples. This division of labor keeps interaction costs low while preserving **human control** and generating **auditable rationale** tied to specific regions of the canvas.

[Figure 4 — Placeholder]

Block diagram: state digest → agent → suggestion cards rendered on the canvas.

5. Footwear as a focused testbed

Footwear design is a strong context for evaluating early exploration tools. The domain combines (a) **structured part vocabularies** that lend themselves to axis and tag names, (b) **concrete briefs** that enable measurable alignment (e.g., traction for wet surfaces, weight targets, seasonal constraints), and (c) **practitioner access** for formative and comparative studies. This specificity supports clear tasks, realistic constraints, and interpretable metrics while remaining representative of broader industrial design workflows.

[Figure 5 — Placeholder]

Labeled shoe diagram with example axes such as “sleek ↔ rugged” and “minimal ↔ embellished.”

6. Positioning and anticipated contribution

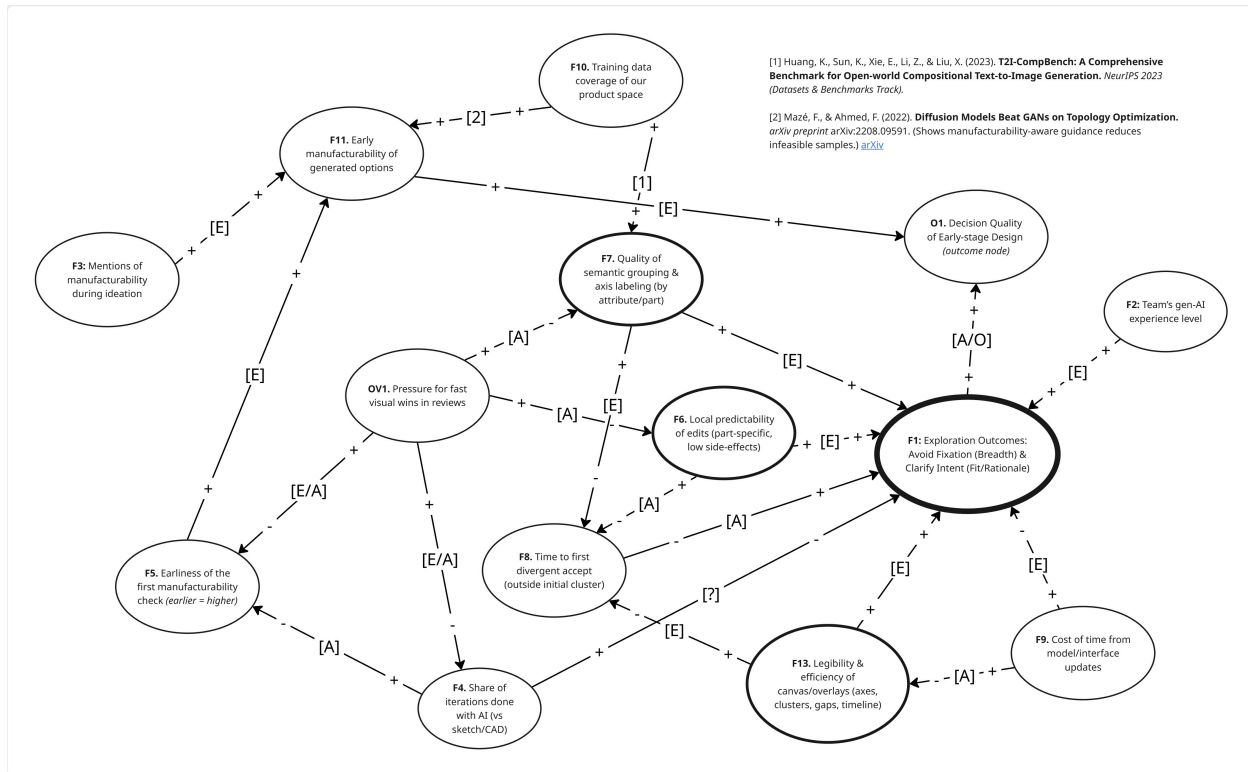
This project integrates three strands: (1) evidence and theory on fixation and exploration breadth, (2) methods for exposing interpretable directions in latent spaces, and (3) interface mediation to address model limits and to maintain alignment with the brief. The expected contributions are an interaction pattern that treats latent space as a **working surface** rather than a hidden substrate, a lightweight **agent protocol** that returns named, on-canvas suggestions with reasons, and **practical measures** for breadth and intent clarity suitable for early-stage evaluation in footwear and related domains.

[Figure 6 — Placeholder]

Table mapping “Challenge → Intervention → Expected signal,” e.g., fixation → gap/contrast highlights → increased switching and coverage.

Models

Reference Model



In many contemporary ideation sessions, especially with text-to-image tools, I observe a consistent pattern: a designer crafts a prompt, generates a grid, selects a small subset of promising images, and then repeats the cycle with slight prompt edits. The system makes it inexpensive to “spin” for more results, so the loop favors volume and quick selection. The trap is that early wins—images with an appealing composition, colorway, or silhouette—anchor subsequent decisions. This anchoring affects not only the look of later generations but also the language of later prompts, which are often edits of the original phrasing. Over time, the process narrows into a path that is visually coherent but rarely surveyed against alternative directions. In the literature, this dynamic aligns with design fixation: early examples bias later choices and reduce exploration breadth (Jansson & Smith, 1991; Sio et al., 2015).

A central reason this narrowing persists is that the **latent structure** of the model remains hidden. Models like CLIP and diffusion systems hold meaningful relationships among images and concepts, but the user mostly sees outputs as disconnected grids. There is no obvious “map” that shows what lies near a chosen result, what lies far away, or which direction one might move to change salient attributes. Without a visible structure, it is difficult to ask, “What are the adjacent possibilities?” or “Where are the unexplored pockets that still fit my brief?” As a result, prompt iteration becomes the only steering wheel, and even careful prompting cannot fully compensate for the lack of spatial awareness.

This absence of structure also undermines **intent clarity**. In early design, intent evolves: a designer may begin with “lightweight trail runner for wet climates,” then realize that drainage, outsole geometry, and upper materials matter more than initial color ideas. In a grid-based loop, that evolution is rarely recorded or externalized. Accepted images are not linked to named reasons, and there is no place on a

map to tag “this is where the water-management direction emerged.” Without such anchoring, it becomes hard to explain why a path is promising or to compare it to other directions in a traceable way. Intent lives in the designer’s head and in scattered prompts rather than in visible structures tied to evidence.

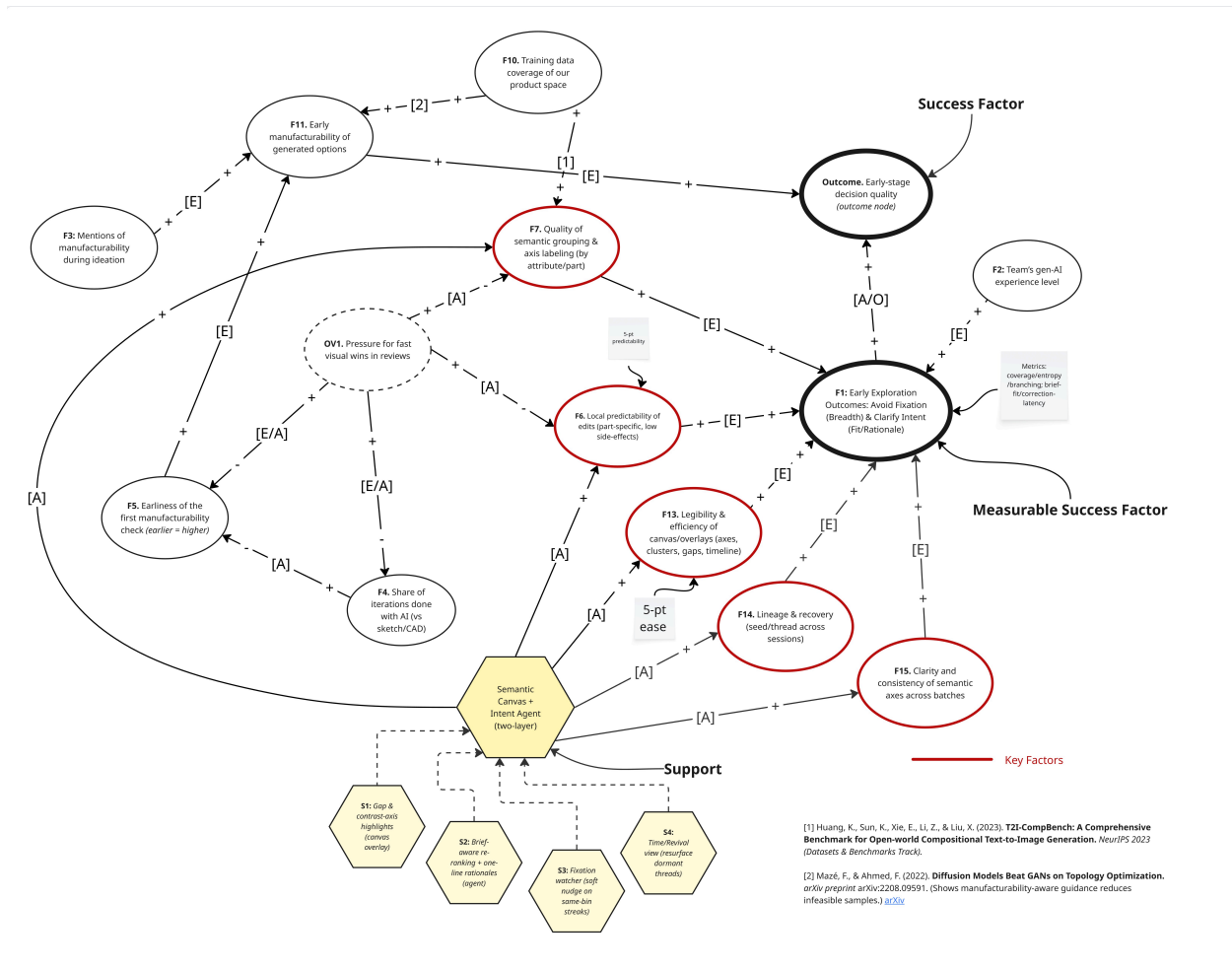
The reference model I propose makes these forces explicit. It starts with a **Brief** that expresses initial intent. A **Prompt-Batch Loop** produces many images quickly. **Early Selection** creates an anchoring effect: the visual and linguistic features of the selected images shape subsequent prompts. The **Hidden Space** is the unexposed latent organization of results; because it is not visible, the designer cannot easily survey alternatives or name directions. This leads to **Narrowing**: exploration contracts around the initial path, and nearby variants increase in fidelity while alternative regions remain unseen. Finally, there is **Intent Drift**: since reasons for choices are not captured in a shared structure (axes, tags, notes tied to regions), the connection between the brief and the evolving set of images becomes fuzzy.

This model is grounded in prior work in two ways. First, the fixation literature provides the psychological mechanism for narrowing (Jansson & Smith, 1991; Sio et al., 2015). Second, research on interpretable latent controls shows that named, manipulable directions exist and can be surfaced (Härkönen et al., 2020; Shen et al., 2020). Together, these bodies of work suggest that the problem is not only in model capability but also in interface design: by failing to expose structure, tools make broad exploration and intent articulation harder than they need to be.

The footwear context sharpens these issues. Footwear brings part vocabularies (upper, tongue, toe cap, heel counter, midsole, outsole), functional constraints (traction, drainage, flexibility), and brand language. Early generative outputs tend to emphasize globally salient appearance features (color, silhouette) over nuanced part-level trade-offs. Without structure, a designer may chase a compelling silhouette while overlooking outsole geometry suited for wet terrain. In other words, the loop rewards seductive images, not necessarily brief-aligned directions.

Summarizing the reference model: **Prompt-Batch** produces volume; **Early Selection** anchors; **Hidden Space** prevents surveying; **Narrowing** increases fidelity within a single path; **Intent Drift** leaves rationale implicit. If I want designers to avoid fixation and clarify intent, the model makes clear where to intervene: make the space visible and nameable; create lightweight ways to record reasons; and provide gentle, well-timed suggestions that point to alternatives without wresting control from the designer.

Impact Model



I introduce two supports. First, a **semantic canvas** places images on a visible 2D/3D map using vision-language embeddings, with named axes and overlays (clusters, gaps, simple lineage/timeline). This gives the designer a sense of “where I am” and “what else is nearby.” Second, a **brief-aware agent** consumes a compact “state digest” (small canvas thumbnail, cluster summaries, and recent actions) and returns a few **small, actionable suggestions** with short reasons: mark a gap to explore, propose axes, surface two or three seed prompts, or highlight relevant prior results. The intent is not to automate design but to counter the narrowing forces and to keep intent visible. The expected outcomes are: (a) avoiding fixation by nudging breadth (coverage across clusters, switching between areas, revisiting alternatives), and (b) clarifying intent by encouraging named axes, tags, and short rationales attached to visible regions of the space.

Research Questions

Main question

How can a semantic canvas and a small, event-driven agent help early-stage footwear designers (a) avoid design fixation and (b) clarify their evolving design intent?

Sub-questions

1. What forms of agent suggestion (for example, highlighting gaps, proposing axes, drafting seed prompts) best increase breadth without overwhelming the designer?
2. Which interaction traces (for example, tags, axis labels, region notes) best support intent clarity during and after exploration?
3. How do outcomes differ across three conditions—baseline prompt-and-batch, canvas only, and canvas + agent?

Hypothesis

Using a semantic canvas plus a compact, suggestion-oriented agent will:

H1. **Reduce fixation:** Designers will cover more distinct regions, switch paths more often, and revisit alternatives more than in baseline prompting.

H2. **Clarify intent:** Designers will report higher intent clarity and leave stronger traces (named axes, tags, notes) that align outcomes with their brief, without increasing time or clicks.

Methods

Prototype status (current)

- **Canvas:** Working 2D and 3D semantic canvas (zoom/pan, axes, early clusters/gaps, basic tagging).
- **Generation backend:** nanobanana integration; embeddings and placement are functional.
- **Agent hook:** Early Gemini-Flash integration can suggest prompts. Planned behaviors: highlight under-explored areas, propose axis labels, provide brief-aware prompts, and “revive this thread.”

Study plan

Stage 1 — Formative feedback

6–10 sessions (45–60 min). Think-aloud exploration and short interviews to refine overlays, suggestion wording, confirmation flows, and timing.

Stage 2 — Controlled comparison (within-subject)

- **Participants:** 10–20 (CMU design students and practicing footwear/industrial designers).
- **Conditions:**
 - **C0:** Prompt-and-Grid baseline.
 - **C1:** Semantic Canvas (no agent).
 - **C2:** Semantic Canvas + Agent (named suggestions with short reasons; accept/ignore).
- **Tasks:** Three short briefs (8–10 min each), counterbalanced across conditions, each yielding 3–5 marked candidates.

Measures

- **Breadth / anti-fixation:** map coverage, variety across accepted regions, number of branches, time to first selection outside the starting area.
- **Intent clarity:** brief match for accepted picks (automatic text-image checks plus brief-specific checklists), time to correct after a miss, use of reasons/labels when accepting.
- **Experience:** time on task, clicks per accepted pick, post-task ratings (predictability, legibility, ease of finding alternatives) and a free-text comment.

Data collected

Interaction logs (prompts, accepts, edits, tags, regions, suggestion views/accepts, timestamps), screen capture, final picks and canvas snapshot, brief surveys, and a short wrap-up interview.

Procedure (Stage 2)

Consent → 3–5 min tutorial → three task blocks (order counterbalanced) → post-task surveys → 5–8 min interview → debrief.

Analysis plan

Within-subject comparisons across C0/C1/C2 for breadth and intent; report per-participant changes and group summaries with effect sizes and confidence intervals. Qualitative coding of screen captures and comments for signs of fixation, moments of intent clarification, and helpful or distracting agent behaviors; refine a small codebook on a pilot set, then code all sessions.

Ethics and IRB

CITI training; IRB submission before recruitment (target early January). Consent covers screen capture and anonymized logs. No personal identifiers in logs; assign IDs; encrypted storage; right to withdraw.

Feasibility and tech choices

The embedding layer for the canvas uses CLIP-like encoders to support semantic placement and axis naming (Radford et al., 2021). The agent uses a fast, low-cost, vision-capable model primarily to prioritize, name, and explain; clustering and scoring remain server-side for cost and stability. I rely on established evidence that latent directions can be made interpretable and controllable (Härkönen et al., 2020; Shen et al., 2020).

Timeline for Completion (draft)

Nov–Dec 2025

- Lock scope of agent actions (gap highlight, axis proposal, revival).
- Polish overlays (clusters/gaps/timeline) and logging.
- Draft IRB, complete CITI training materials.

Early Jan 2026

- **Submit IRB;** prepare recruitment materials and brief templates.
- Internal pilot (3–5 users) → refine suggestion wording and triggers.

Feb 2026

- Run Stage 1 formative sessions (6–10).
- Update UI/agent based on findings; finalize study protocol.

Mar 2026

- Run Stage 2 within-subject study ($N \approx 10-20$).
- Ongoing data cleaning; begin qualitative coding.

Apr 2026

- Quant + qual analysis; compute breadth/intent metrics.
- Draft Results, Discussion, and Limitations.

May 2026

- Final synthesis; figures/tables; proofread.
- Submit thesis package; prepare presentation and, if viable, a short paper.

Results to Date / Anticipated Findings

So far. I have a functional canvas (2D/3D) with image generation integrated, and early agent scaffolding for prompt suggestions. Two exploratory interviews with footwear designers confirmed the value of seeing the “space” and receiving gentle nudges away from local maxima.

What I expect to find. Compared to baseline prompting, Canvas should increase breadth; Canvas + Agent should further expand breadth and improve intent clarity and rationale quality. Compositional limits in today’s text-to-image models (for example, attribute binding and relationships) may still appear; I expect the agent to help route exploration around such pitfalls by steering toward clearer regions or reframing prompts (Huang et al., 2023).

Discussion / Contributions

- **Practical pattern for agent-supported exploration:** A light **state-digest** → **suggestion JSON** loop that enhances exploration without flooding the UI.
- **Legible latent-space interaction:** Named axes, gap highlights, and linked notes to make model behavior and design intent easier to see and explain.
- **Breadth/clarity metrics for early-stage work:** Simple, reusable measures (coverage, switching, rationale alignment).
- **Domain case with generalizable lessons:** Insights from footwear that likely transfer to other industrial design contexts.

Limitations and Next Steps

- **Model limits and bias:** Compositional errors remain a challenge; the system can only steer around some of them (Huang et al., 2023).

- **Scope and sample:** Early studies (N≈10–20) at one university and one industry partner limit generality.
- **Task realism:** Short sessions may not capture collaboration, cost, materials, or manufacturing constraints.

Next steps. Finalize IRB; complete agent actions beyond prompt suggestion (gap highlighting, axis proposals, gentle nudge, revival); add coverage/branching/timeline overlays and dashboards; pilot (3–5 users); run the counterbalanced study; pursue industry integration and publication.

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